Team Assignment

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Graph Theory / Combinatorics

Named Problem: Assignment Problem

Universe

- Assignment: \{Alice, Bob, Carol, Dave, Eve, Frank\}

Variables

Symbol	Meaning	Type / Range
x_{ij}	1 if item i assigned to slot j, 0 otherwise	\{0, 1\}

Structure

Assign 6 developers to 6 projects (one developer per project, one project per developer). Each developer has rated their interest in each project on a scale of 1-10. Find the assignment that maximizes total satisfaction across all developer-project pairs.

Real-World Mapping

Real-World Concept	Mathematical Object
assignment	x: solution vector (assignment)
total score/cost	objective function value

Constraints

(1)

$$\sum_j x_{ij} = 1$$

one-per-item

(2)

$$x_{ij} \in \{0, 1\} \quad \forall i, j$$

binary decision variables

Objective

$$\max \sum_{i,j} c_{ij} x_{ij}$$

Approach

Suggested approach: Hungarian Algorithm (scipy.optimize.linear_sum_assignment)

Complexity class:	O(n^3)
Available tools:	Hungarian Algorithm

2. Solution

Answer:	6-element assignment: Alice -> ML Model, Bob -> Mobile App, Carol -> Da
Objective Value:	52
Optimal:	Yes
Feasible:	Yes
Algorithm:	Hungarian Algorithm (scipy.optimize.linear_sum_assignment) O(n^3)
Time:	0.0001s
Certificate:	Hungarian algorithm guarantees global optimum in O(n^3).

Solution Details

Assignment:
  Alice -> ML Model
  Bob -> Mobile App
  Carol -> Data Pipeline
  Dave -> DevOps CI/CD
  Eve -> API Gateway
  Frank -> Frontend UI

Objective value: z* = 52
Upper bound: 53
Lower bound: 16

Random Stats:
  mean: 33.8905
  std: 5.6285
  min: 18
  max: 52
  p95: 43.0000

Method: Hungarian algorithm (Kuhn-Munkres) via scipy.optimize.linear_sum_assignment. Since the algorithm minimizes cost, we negate the preference matrix to convert maximization into minimization. The algorithm runs in O(n^3) and guarantees a globally optimal assignment.

Verification

✓	Feasibility	All constraints satisfied

✓	Optimality	Hungarian Algorithm (scipy.optimize.linear_sum_assignment)

3. Interpretation

The Question

Assign 6 developers to 6 projects (one developer per project, one project per developer).

The Answer

The optimal solution has objective value z* = 52.

What This Means

- Optimal one-to-one assignment of developers to projects
- Maximum total satisfaction score
- Comparison against worst-case and random assignments

Hungarian algorithm (Kuhn-Munkres) via scipy.optimize.linear_sum_assignment. Since the algorithm minimizes cost, we negate the preference matrix to convert maximization into minimization. The algorithm runs in O(n^3) and guarantees a globally optimal assignment.

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio